Haroon Rashid

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About Me

Computer Scientist and Computer Vision Engineer with close to 3 years of experience in 2D Computer Vision and Python. 20+ high-quality projects made, 150+ models trained, and 3 high-quality datasets annotated..

Currently, a research scientist in the field of 3D Computer Vision.

Education

National University Of Modern Languages, Islamabad, BS in Computer Science — Jan 2023 – Jan 2027

- GPA: 3.87/4.0

Projects

Tennis AI — webpage

- Utilized 2D Computer Vision techniques, including but not limited to region-based object detection, object tracking, object speed estimation, and multiple perspective transformations, for a detailed tennis match analysis.
- Accurately estimated ball speeds, classified players, and produced a radar view of the court using multiple perspective transformations.
- Tools Used: Python, Ultralytics, Supervision, OpenCV, Numpy.
- Model(s) Used: YOLOv8x and YOLOv10x.

Volleyball AI — webpage

- Utilized 2D Computer Vision techniques, including but not limited to object detection, key point regression, perspective transformations, and homography calculations, for producing a radar view of a volleyball match recorded by a non-static camera.
- Precisely annotated three datasets, trained three YOLOv8 models, and produced a radar view of the volleyball match.
- Tools Used: Python, Ultralytics, Supervision, OpenCV, Numpy.
- Model(s) Used: YOLOv8x-pose and YOLOv8x.

Destruction Estimation Using Segmentation and SAHI — webpage

- Utilized 2D Computer Vision techniques, including but not limited to, semantic segmentation and SAHI (slicing-aided-hyper-inference), to estimate the destruction caused to construction in an area. I have accurately estimated the number of buildings lost due to destruction..
- Tools Used: Python, Ultralytics, Supervision, OpenCV, Numpy.
- Model(s) Used: YOLOv8x

Mathematical Algorithm Implementations — webpage

- Implemented several mathematical algorithms for solving systems of linear equations and for approximating the roots of non-linear equations, in pure Python.
- Tools Used: Python

Technologies

Languages: Python, C++, Java, LaTeX, MySQL

Skills: 2D Computer Vision, 3D Computer Vision (research), OpenVINO, PyTorch, Keras, PyTorch3D, Computer Vision Data Annotation, YOLO, Linux, Algorithm Design, Mathematics, Python Desktop and Web App Development

Tools: PyCharm, Kaggle Notebooks, Spyder, IDLE, MongoDB Compass, XAMPP, WANDB, Roboflow Workspace, CVAT